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ABSTRACT

Stock Price Trend Analysis using Power BI is a data visualization project designed to study historical stock market movements and identify patterns in stock prices over time. Investors and analysts often need to examine large amounts of historical market data to understand price fluctuations, compare performance and identify periods of growth or decline. Reading raw stock data alone can make these patterns difficult to understand.

This project uses Microsoft Power BI to import, clean, transform and visualize historical stock price data. Line charts are used to display price movements over time, while area charts highlight the magnitude of price changes. A column chart compares the average previous closing price across years, and DAX measures such as average closing price, highest price, lowest price, first and latest closing price, price change and percentage change help summarize stock performance.

The resulting dashboard allows users to filter the stock data by date range, explore price movements and compare trends. The project demonstrates the practical use of data cleaning, calculated measures, interactive visualizations and dashboard design for financial data analysis.

Keywords: Power BI, Stock Market, Data Visualization, Line Chart, Area Chart, Historical Stock Prices, Trend Analysis, Financial Analytics.

1. INTRODUCTION

The stock market provides a platform for buying and selling shares of publicly listed companies. Stock prices change over time due to several factors, including company performance, economic conditions, market demand, investor confidence and global events. Every trading session produces financial information such as opening price, highest price, lowest price, closing price and trading volume. These values provide information about the price movements and trading activity of a stock during a particular period.

Historical stock market data is useful for understanding how a stock has performed over time. By examining closing prices across different dates, users can observe increases, decreases and fluctuations in the stock's value. Similarly, the highest and lowest prices provide information about the price range recorded during a trading session, while trading volume indicates the number of shares traded. When these attributes are analysed together, they provide a broader view of historical market activity.

However, stock market datasets may contain hundreds or thousands of records, making manual analysis difficult. Raw tables provide individual values but do not always make the overall trend easy to understand. Users may need to sort records, perform calculations, create charts and compare results separately. This process can become time-consuming, particularly when different date ranges or several stocks need to be examined.

Data visualization tools help solve this problem by converting numerical information into graphical representations. Microsoft Power BI is a business intelligence platform that allows users to import data, transform it, perform calculations and create interactive reports. It provides several visualization options, including line charts, area charts, column charts and card visuals. These features help users understand complex information through a structured and user-friendly interface.

1.1 Objectives of the Project

The primary objective of this project is to develop an interactive Power BI dashboard for analysing historical stock market data. The project aims to collect and prepare stock data, examine its important attributes, calculate relevant indicators and present the results through suitable visualizations. It also aims to reduce the effort required to understand raw numerical records by providing charts and summary cards in a single report.

Another important objective is to understand the practical use of Power Query and DAX in data analysis. Power Query helps prepare the dataset, while DAX measures perform calculations that respond to the selected filters. The project also aims to provide interactive filtering, improve the presentation of financial information and validate the displayed results against the original dataset. Through these activities, the project demonstrates how Power BI can transform historical stock records into meaningful visual information.

2. EXISTING SYSTEM AND PROPOSED SYSTEM

2.1 Existing System

In the traditional approach, historical stock market data is commonly analysed using spreadsheets, raw tables or manually prepared charts. Users download the dataset, open it in a spreadsheet application, examine the records and calculate summary values using formulas. Separate charts may then be created to display closing prices and trading volume. Although this method can be useful for small datasets, it becomes more difficult when the number of records increases or when several indicators need to be analysed together.

Manual analysis requires users to spend time organizing data, correcting errors, updating formulas and preparing visual representations. When a different date range needs to be examined, the user may have to modify the calculations and charts again. Static reports can also limit interaction because users may need to prepare separate views to examine different stocks or periods. Furthermore, displaying important values across multiple tables and charts can make the overall analysis less convenient.

The existing approach therefore has limitations in terms of time, interactivity and presentation. Although spreadsheets can perform calculations and create charts, a structured dashboard provides a more convenient way to combine different indicators and explore historical data.

2.2 Proposed System

The proposed system uses Microsoft Power BI Desktop to analyse historical stock market data and display the results through an interactive dashboard. The system begins by importing a stock dataset from a CSV file or another supported source. The imported records are checked and transformed using Power Query to ensure that the date and numerical columns are suitable for analysis. Missing values, duplicate records and inconsistent data types are examined before the dataset is loaded into the Power BI model.

After data preparation, DAX measures are created to calculate indicators such as average closing price, highest price, lowest price, latest closing price and percentage price change. These calculations are then used in card visuals and charts. A line chart presents the historical closing-price trend, an area chart provides another view of price movement, a column chart displays trading volume and a year-wise column chart compares the average previous closing price across years. The visualizations are arranged in a single report page so that users can examine the information conveniently.

An interactive slicer is added to filter the data by date range. When the user changes the selected range, the relevant charts and measures respond according to the selected data. This reduces the need to prepare separate reports for different periods and makes the dashboard easier to use.

The proposed system improves the presentation of historical stock information by combining data preparation, calculations and visualization in one environment. It also makes the analysis easier to review and update when new data is loaded. The system is intended for historical analysis and does not provide investment recommendations or future price predictions.

Table 2.1 summarizes the differences between the existing and the proposed systems.

Table 2.1: Comparison of the Existing and Proposed Systems

Feature	Existing System	Proposed System
Data analysis	Manual analysis using spreadsheets and raw tables	Analysis using Power BI
Data preparation	Manual correction and organization	Power Query transformations
Calculations	Manually entered formulas	Reusable DAX measures
Visualization	Separate charts or static reports	Interactive dashboard
Filtering	May require separate views	Interactive date-range slicer
Updating	Charts and formulas may need manual changes	Prepared steps and measures can be reused when data is refreshed
Result presentation	Data displayed across separate tables and charts	Charts and KPI cards combined in one report

3. TOOLS, TECHNOLOGIES AND DATASET DESCRIPTION

3.1 Tools and Technologies Used

Microsoft Power BI Desktop is the main application used to develop the project. It provides the environment for importing data, creating measures, preparing visualizations and designing the final dashboard. Its interactive features allow users to filter and examine the data without creating a separate report for every selection.

Power Query is used for data preparation and transformation. It allows the user to inspect the dataset, change column data types, remove unnecessary fields and examine missing or inconsistent values. The transformations are recorded as steps, making the preparation process easier to review and repeat when the source data is refreshed.

Data Analysis Expressions, commonly known as DAX, is the formula language used to create calculations in Power BI. DAX measures are used to calculate important stock indicators based on the available data and the current filter context. For example, the highest-price measure returns the maximum value in the High column for the selected records, while the price-change measure calculates the difference between the first and latest closing prices in the selected period.

The project uses a historical stock market dataset, preferably in CSV format. A CSV file stores data in a tabular structure, where each row represents a record and each column represents an attribute. The dataset should contain the fields required for analysis, such as Date, Open, High, Low, Close, Previous Close and Volume. A Symbol column may also be available when the dataset contains multiple stocks.

3.2 Dataset Description

The dataset forms the foundation of the entire project because the accuracy of the analysis depends on the quality and completeness of the source records. Historical stock data usually contains daily information about the prices and trading activity of a selected company. The Date column identifies the trading session, while Open, High, Low and Close describe different price values recorded during that session. The Volume column represents the number of shares traded.

The Date attribute is used to arrange the records chronologically and create time-based visualizations. Open represents the opening price, High represents the highest recorded price, Low represents the lowest recorded price and Close represents the closing price. Prev Close represents the closing price of the preceding trading session and is used in the year-wise comparison chart. The closing price is the primary field used to display historical price trends, and Volume is used to examine variations in trading activity. The dataset also contains the Symbol, Series, Last, VWAP, Turnover, Trades, Deliverable Volume and %Deliverble columns, which are not used in the price-trend analysis. The dataset does not contain an Adjusted Close column, so the prices are not adjusted for corporate actions such as stock splits.

The dataset was examined to confirm that the required columns were present and that the records covered a clearly defined period. The dates are in chronological order, and the file has no duplicate records. The Trades column is empty for the period before June 2011, and the Deliverable Volume and %Deliverble columns have 509 missing values; none of these columns is used in the analysis, so the missing values do not affect the results. The details of the dataset are given below.

- Dataset name: Asian Paints Historical Stock Price Data (archive-1.csv)
- Company or stock symbol: Asian Paints Limited (ASIANPAINT), series EQ
- Data source: NIFTY-50 Stock Market Data (2000–2021), Kaggle, based on National Stock Exchange of India (NSE) data
- Date range: 3 January 2000 to 30 April 2021
- Number of records: 5,306 daily records
- File format: CSV, with 15 columns

3.3 System Requirements

The project requires a computer or laptop capable of running Microsoft Power BI Desktop, sufficient memory to load the selected dataset and adequate storage space for the application and project files. An internet connection may be required to download the historical stock data and access official documentation.

The main software requirement is Microsoft Power BI Desktop. Power Query is used for data preparation, and DAX is used for calculations. The dataset can be stored in CSV format or another format supported by Power BI. A word processor is used to prepare the final project report and insert screenshots of the completed implementation.

4. METHODOLOGY AND SYSTEM IMPLEMENTATION

The project follows a systematic process that converts raw stock market records into an interactive dashboard. The major stages are data collection, data import, data cleaning, data transformation, calculation of indicators, visualization, dashboard development and testing. Each stage is important because incorrect data preparation or calculation can affect the final results.

4.1 Data Collection, Import and Preparation

The first step is to collect historical stock market data from a suitable source. The selected dataset should contain the required price and volume attributes for a clearly identified stock and date range. After downloading the file, the source details are recorded so that the data can be referenced in the final report.

The dataset is imported into Power BI Desktop using the Get Data option. If the file is stored in CSV format, the Text/CSV connector is selected. The preview is checked to confirm that the columns and values are displayed correctly. Transform Data is then selected to open Power Query Editor, where the dataset is prepared before loading it into the report.

During data preparation, the Date column is assigned the Date data type, and the price columns are converted to suitable numeric types. The Volume column is checked to ensure that its values can be used for numerical analysis. Missing values, duplicate records and unnecessary columns are reviewed. Any record that appears incomplete should be handled according to the source data and a consistent cleaning rule rather than being removed without verification.

The column names and data structure are also examined to ensure that the dataset is easy to understand. The dataset contains a single stock (ASIANPAINT), so the Symbol column is not used for filtering. After the required transformations are complete, Close & Apply is selected to load the prepared data into the Power BI model.

4.2 Creating DAX Measures

After the data has been loaded, DAX measures are created to calculate the important indicators required for the dashboard. These measures allow the report to display summary values without requiring the user to calculate them manually each time. The following formulas assume that the table is named StockData and contains the columns Date, Close, High and Low. If the actual names are different, the formulas must be adjusted accordingly.

The average closing price is calculated using the following measure:

Average Closing Price

```
Average Closing Price =
AVERAGE(StockData[Close])
```

This measure calculates the average of the closing-price values within the current filter context. It can be used to summarize the average closing price for the selected period.

The highest and lowest recorded prices are calculated using the following measures:

Highest and Lowest Prices

```
Highest Price =
MAX(StockData[High])

Lowest Price =
MIN(StockData[Low])
```

The Highest Price measure returns the maximum value from the High column, while the Lowest Price measure returns the minimum value from the Low column. These measures help summarize the recorded price range within the selected data.

To compare the first and latest closing prices, two additional measures are created:

First Closing Price

```
First Closing Price =
VAR FirstDate =
    MINX(
        ALLSELECTED(StockData[Date]),
        StockData[Date]
    )
RETURN
    CALCULATE(
        MAX(StockData[Close]),
        StockData[Date] = FirstDate
    )
```

Latest Closing Price

```
Latest Closing Price =
VAR LatestDate =
    MAXX(
        ALLSELECTED(StockData[Date]),
        StockData[Date]
    )
RETURN
    CALCULATE(
        MAX(StockData[Close]),
        StockData[Date] = LatestDate
    )
```

The first measure identifies the earliest selected date and returns its closing price. The second identifies the latest selected date and returns its closing price. These examples assume one applicable closing-price value per date within the relevant stock context. When multiple stock symbols are present, the appropriate stock filter must be retained.

The difference between the latest and first closing prices, and the same difference as a percentage of the first closing price, are calculated as follows:

Price Change

```
Price Change =
[Latest Closing Price] - [First Closing Price]
```

The Price Change measure shows the difference between the latest and first closing prices. A positive value means that the latest closing price is higher than the first closing price, while a negative value means that it is lower. The Price Change % measure expresses this difference relative to the first closing price; the DIVIDE function returns a blank instead of an error if the first closing price is zero. The measure should be formatted as a percentage in Power BI.

These measures summarize historical values for the selected period. They do not predict future stock prices, and their correctness should be verified against the underlying dataset before the results are reported.

4.3 Creating Visualizations and the Dashboard

After the required measures have been created, the next step is to build the visualizations. The line chart is the main visual because it displays how the closing price changes over time. The Date field is placed on the X-axis, and the Close field or an appropriate closing-price measure is placed on the Y-axis. The chart is titled Historical Stock Closing Price Trend, and the date axis is checked to ensure chronological ordering.

An area chart is created to provide another view of historical closing-price movements. The Date field is placed on the X-axis, and the closing-price field is placed on the Y-axis. The chart is titled Stock Price Movement Over Time. This visual complements the line chart by presenting the same price series using an area representation.

A clustered column chart is used to display trading volume. The Date field is placed on the X-axis, and the Volume field is placed on the Y-axis. The aggregation must match the structure of the dataset. If the data contains one daily volume record per stock, the daily value should be represented correctly. The chart is titled Trading Volume Over Time and allows users to compare trading activity across the selected period.

A second clustered column chart is created for the year-wise comparison. The Year (derived from the Date field) is placed on the X-axis, and the average of the Prev Close field is placed on the Y-axis. The chart is titled Average of Prev Close by Year and allows users to compare annual values (Figure 5.1).

Card visuals are added to summarize important indicators. Separate cards can display the latest closing price, highest price, lowest price and percentage price change. Each card is assigned the appropriate measure, and the number format is adjusted to make the values easy to read. Price measures should use consistent decimal formatting, while the percentage measure should be displayed as a percentage.

An interactive slicer is added to allow users to examine selected date ranges. The slicer is tested to ensure that the charts and cards respond correctly to the selected values.

Finally, all visuals are arranged into a single dashboard page. The project title and KPI cards are positioned near the top, followed by the main line chart and supporting visualizations. The area chart, trading-volume chart and year-wise chart are arranged so that the page remains clear and readable. Consistent titles, suitable axis labels, appropriate number formats and adequate spacing are applied to improve the overall presentation.

4.4 Key Features of the Dashboard

The Stock Price Trend Analysis dashboard developed using Microsoft Power BI provides several features to simplify the analysis of historical stock market data.

Historical Price Analysis: The dashboard displays stock price movements over time, helping users observe changes in prices across the selected period.

Year-wise Price Comparison: A column chart displays the average previous closing price for each year, allowing users to compare annual values and identify variations over time.

Summary Measures: DAX measures calculate important indicators such as average closing price, highest price, lowest price, first closing price, latest closing price and percentage price change.

Interactive Visualizations: Charts and other visuals present the selected stock data in a clear and organized format, making historical trends easier to understand.

Data Preparation: Power Query is used to prepare the dataset by checking column data types, reviewing missing values and correcting inconsistencies where required.

Data Filtering: A date-range slicer can be used to examine specific periods in the data.

These features combine data preparation, calculations and visual representation in a single report. The dashboard helps users understand historical stock-price behaviour and compare values more conveniently than examining raw data alone.

5. RESULTS AND DISCUSSION

The Stock Price Trend Analysis project was developed using Microsoft Power BI to analyse historical stock market data and present the results through visualizations. The dataset was imported and prepared using Power Query, and DAX measures were used to calculate important stock price indicators. The dashboard helps users understand historical price movements, compare values across different periods and examine changes in trading activity.

The analysis was performed using the stock dataset for Asian Paints Limited (ASIANPAINT), covering the period from 3 January 2000 to 30 April 2021. The dataset contains 5,306 records.

5.1 Power BI Dashboard for Stock Price Trend Analysis

The Power BI dashboard presents the historical stock price data using charts, KPI cards and slicers. The line chart displays changes in the closing price over time, the area chart gives an alternative view of the same movement, and the trading-volume chart shows variations in trading activity. The KPI cards display the highest price, lowest price, latest closing price and percentage price change. The year-wise column chart compares the average previous closing price for each year, as shown in Figure 5.1.

The date-range slicer allows users to examine selected periods and observe how the displayed values change according to the selected range.

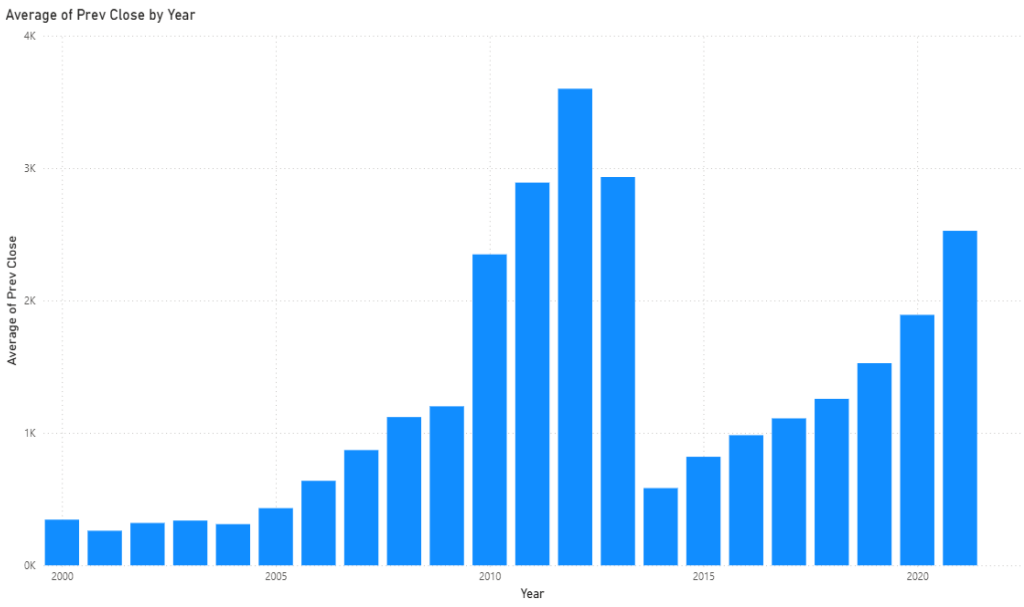


Figure 5.1: Year-wise Average Previous Closing Price (Power BI)

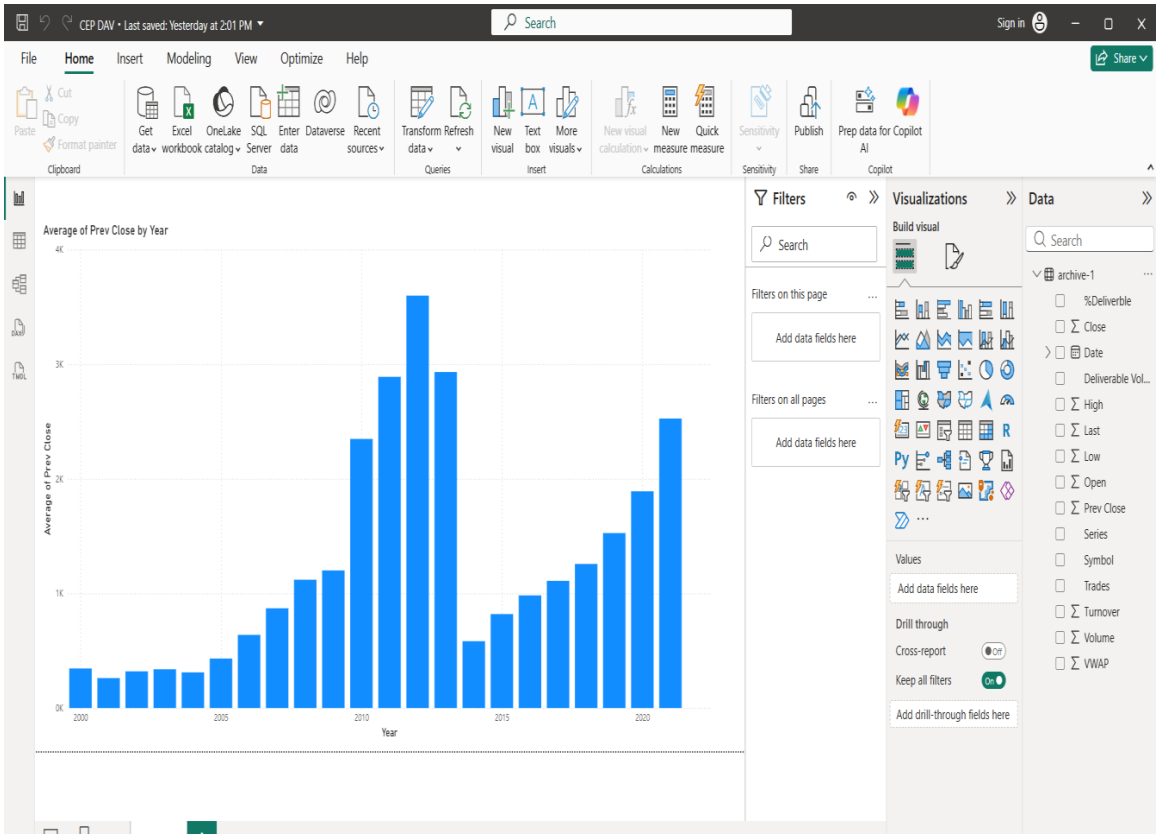


Figure 5.2: Complete Power BI Dashboard for Stock Price Trend Analysis

5.2 Analysis of Stock Price Trends

The stock price trend is analysed by observing the changes shown in the visualizations. In the year-wise comparison chart (Figure 5.1), the average previous closing price remains low and fairly stable between 2000 and 2005, then rises steadily from 2006 and sharply between 2010 and 2012, reaching a peak of about ₹3,599 in 2012. The value then falls to about ₹583 in 2014. This fall does not represent a collapse in the company's value: the dataset contains unadjusted prices, and the closing price dropped from ₹5,118.20 to ₹510.75 on 30 July 2013, which is consistent with a 10-for-1 stock split. After the split, the average rises gradually every year to about ₹2,526 in 2021, which covers only January to April.

The highest and lowest price measures summarize the recorded price range. The first and latest closing prices are compared to understand the overall change during the selected period, and the percentage price change provides a numerical representation of this difference.

Trading volume is examined to understand how the number of shares traded varies over time. The results should be interpreted based on the actual values and patterns displayed in the dashboard.

5.3 Summary of Results

The dashboard provides a visual summary of historical stock market data and makes it easier to examine price movements than reviewing raw records alone. Power BI charts support comparisons across time periods, while DAX measures summarize important values from the dataset.

The main results obtained from the analysis are listed below:

- Company analysed: Asian Paints Limited (ASIANPAINT)
- Date range analysed: 3 January 2000 to 30 April 2021
- First closing price: ₹381.65 (3 January 2000)
- Latest closing price: ₹2,536.40 (30 April 2021)
- Price change: ₹2,154.75
- Percentage price change: 564.59%
- Highest recorded price: ₹5,247.75 (19 July 2013, before the stock split)
- Lowest recorded price: ₹204.00 (4 October 2000)

These results provide a descriptive overview of the stock's historical performance. Because the prices in the dataset are not adjusted for the 2013 stock split, the highest recorded price is higher than the post-split prices and should not be compared directly with the later values. The accuracy of the findings also depends on the quality of the dataset, the correctness of the DAX calculations and the proper configuration of the visualizations. Historical price movements describe past performance and do not guarantee future market trends.

6. TESTING AND VALIDATION

Testing is performed to verify the accuracy of the imported data, DAX calculations, visualizations and interactive filters. The imported dataset is compared with the original file to confirm that the expected columns and records are available. The Date column is checked to ensure that it uses an appropriate date data type, while the price and volume columns are checked to confirm that they can be used in numerical calculations.

The DAX measures are validated against the source data. The Highest Price measure should match the maximum value in the High column for the selected stock and date range, while the Lowest Price measure should match the minimum value in the Low column. The Price Change measure should equal the latest closing price minus the first closing price. The percentage change should match the result obtained by dividing the price difference by the first closing price, provided that the starting value is valid.

The visualizations are also checked to ensure that they display the intended fields and use suitable aggregations. The line chart and area chart should show closing-price movements in chronological order, while the volume chart should represent trading activity correctly. The date slicer is tested by selecting different date ranges and observing whether the related visuals update as expected.

Finally, the dashboard is reviewed for presentation issues such as overlapping charts, unclear titles, unsuitable number formats and unreadable labels. The report is finalized only after the calculations, chart configurations and interactive filters have been checked. These validation steps improve the reliability of the dashboard and help ensure that the conclusions are based on the actual data.

Test Area	What Was Checked	Expected Result
Data import	Imported dataset compared with the original file	Expected columns and records are available
Date column	Data type of the Date column checked	Uses an appropriate date data type
Price and volume columns	Columns checked for use in calculations	Can be used in numerical calculations
Highest Price measure	Compared with the maximum of the High column	Matches the maximum High value for the selected stock and date range
Lowest Price measure	Compared with the minimum of the Low column	Matches the minimum Low value for the selected stock and date range
Price Change measure	Compared with latest closing price minus first closing price	Equals the difference between the two closing prices
Price Change % measure	Compared with price difference divided by first closing price	Matches the calculated result, provided the starting value is valid
Line and area charts	Order of closing-price values checked	Closing-price movements shown in chronological order
Volume chart	Trading-activity values checked	Trading activity represented correctly
Date range slicer	Different date ranges selected	Related visuals update as expected
Dashboard layout	Reviewed for presentation issues	No overlapping charts, unclear titles, unsuitable number formats or unreadable labels

7. APPLICATIONS AND ADVANTAGES

The project can be used for historical stock performance analysis, financial data visualization and educational purposes. It provides a convenient way to examine how closing prices have changed over a selected period and how trading volume varies across trading sessions. The same approach can be applied to other stocks by loading their historical data into the dashboard.

One of the main advantages of the project is that it presents important information through a structured visual report instead of requiring users to examine every row of the original dataset. The combination of charts and KPI cards provides an organized summary of the selected data. The interactive date slicer allows users to explore different periods, reducing the need to prepare separate reports for each selection.

The use of Power Query and DAX also improves the organization of the analysis. Data preparation steps can be reviewed and repeated when the source is refreshed, while reusable measures help maintain consistency across visualizations. The project provides practical experience in data cleaning, calculations, visualization and dashboard design. These skills are applicable to other areas of business intelligence, including sales analysis, financial reporting and performance monitoring.

The dashboard is intended to support the understanding of historical information. Its usefulness depends on the quality of the source dataset and the correctness of the implemented calculations.

8. LIMITATIONS AND FUTURE ENHANCEMENTS

The project has certain limitations because the analysis depends on the quality, completeness and date range of the selected stock dataset. Missing or incorrect records can affect the calculated indicators, while a dataset containing only one stock cannot support comparisons across several companies. The dashboard also depends on the fields provided by the source, so additional analysis may not be possible if the required attributes are unavailable. The prices in the dataset are also not adjusted for corporate actions such as the 2013 stock split, so long-term price comparisons must be interpreted with care.

Another limitation is that historical data does not guarantee future market performance. Stock prices are influenced by many factors, including company announcements, economic developments, market conditions and investor behaviour. The project does not automatically explain these causes, and its visualizations should not be interpreted as financial predictions. If updated data is required, the source file must be refreshed or connected to a suitable data source that supports updating.

The project can be enhanced by including additional stock symbols and longer historical periods. This would allow users to compare the performance of different stocks over a common date range. Additional indicators, such as daily price change, opening-price measures and moving averages, could also be introduced to provide more detailed analysis. Moving averages may help present a smoothed view of historical price movements, provided the calculation window is clearly defined.

Further improvements could include additional dashboard pages, tooltips, drill-through functionality and an appropriate data-refresh mechanism. These features would make the report more informative and easier to navigate. More advanced analytical techniques could also be explored as a separate extension, but they would require suitable data, testing and a clear explanation of their assumptions.

9. CONCLUSION

The project Stock Price Trend Analysis using Power BI demonstrates how business intelligence techniques can be applied to historical stock market data. The main purpose of the project is to convert raw numerical records into an organized and interactive dashboard that presents stock-price movements, trading activity and important summary indicators.

The project involves collecting and importing historical stock data, preparing the records using Power Query and creating DAX measures for calculating relevant indicators. Line charts, area charts, column charts and KPI cards are used to represent the information in a clear format. An interactive slicer allows users to examine selected date ranges.

The dashboard provides a convenient way to study historical closing prices, review the highest and lowest recorded prices, compare the first and latest closing prices and observe variations in trading volume. Testing and validation are important parts of the implementation because the accuracy of the final results depends on correct data preparation, appropriate calculations and properly configured visualizations.

Through this project, practical knowledge is gained in data transformation, DAX calculations, data visualization and interactive report development. It demonstrates that Power BI can be used to transform historical financial records into meaningful visual information. The completed dashboard supports descriptive analysis of the selected stock data, while future market performance remains uncertain and cannot be guaranteed by historical trends alone.

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